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Virtual Gold (AI Assistant) x CMU Capstone: Weekly Sync

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Attachments  Virtual Gold (AI Assistant) x CMU Capstone: Weekly Sync

Summary

Project scope discussion via vertical scenario evaluation and horizontal architectural planning.

Project Scope And Scenarios

Staging the horizontal scenario first proved most effective. Focusing on a chief of staff assistant framework streamlined project direction.

Security Research And Architecture

Security research highlighted sensitive data handling strategies. A proposed security architecture required redesigning for the horizontal scenario.

System Integration And Deliverables

Building a standalone server with an HTTP endpoint established testing protocols. Creating an initial architecture diagram delivered the required next milestone.

Decisions

Aligned

- **Prioritization of horizontal use case scope** The team aligned to prioritize developing the horizontal scenario and privacy module first, deferring the vertical scenario to time-permitting status.
- **Framing of horizontal IT department use case** The participants agreed to frame the horizontal application around an IT department capability supporting a chief of staff assistant.
- **HTTP API endpoint integration architecture** The team aligned to develop the hybrid local model module as a standalone service with an HTTP API endpoint for integration.
- **Horizontal architecture draft for next meeting** The team aligned to build and share a draft of the horizontal plot architecture and system diagram as the deliverable for the next meeting.

Next steps

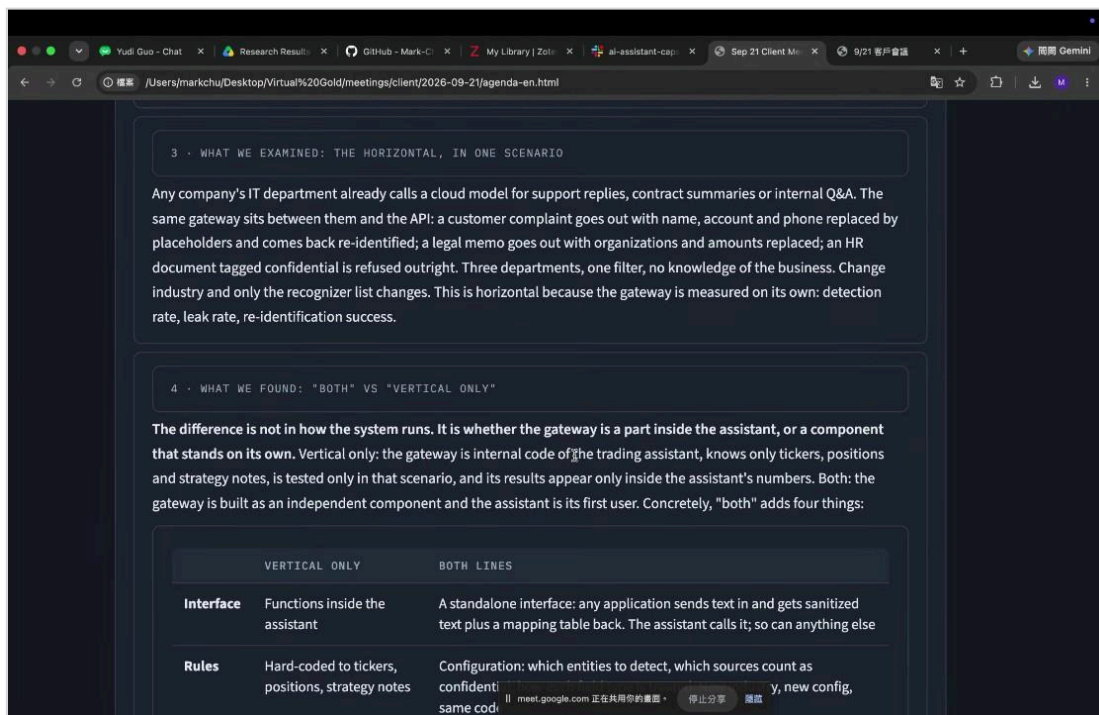
- ☐ [Mark Chu] Update Meeting Label: Mention AI hybrid assistant when posting future meetings to ensure correct identification.
- ☐ [Mark Chu] Resend Meeting Documents: Distribute the agenda and discussion documents to all participants.
- ☐ [Inderpal Bhandari] Follow Up NDA: Confirm the status of the NDA addendum with Diane at CMU.
- ☐ [Alex, Aarvin George] Define Endpoint Specifications: Detail the request parameters and expected response format for the HTTP endpoint to enable integration.
- ☐ [The group] Design System Architecture: Draft the architecture for the horizontal privacy module and its integration components.
- ☐ [The group] Define Success Conditions: Establish ground truth and performance metrics to evaluate the assistant.
- ☐ [The group] Develop Golden Data Suite: Create a dataset for regression testing to ensure model performance consistency.

- ☐ [Aarvin George] Provide API Contract: Submit the initial API contract by tomorrow. Coordinate with the team to adapt the definition as requirements evolve.
- ☐ [Mark Chu] Share Google Drive: Configure the virtual gold Google Drive and provide access to Aarvin, Alex, Inderpal, and Alra.
- ☐ [Mark Chu] Setup New Slack: Create a separate Slack group for the capstone to test if the trial limitations persist and share the invitation link with everyone.
- ☐ [The group] Design Architecture: Develop the full system architecture and diagram focusing on the horizontal component. Present this material for review during the next meeting.
- ☐ [Mark Chu] Rename Project: Change the project name to Hybrid.

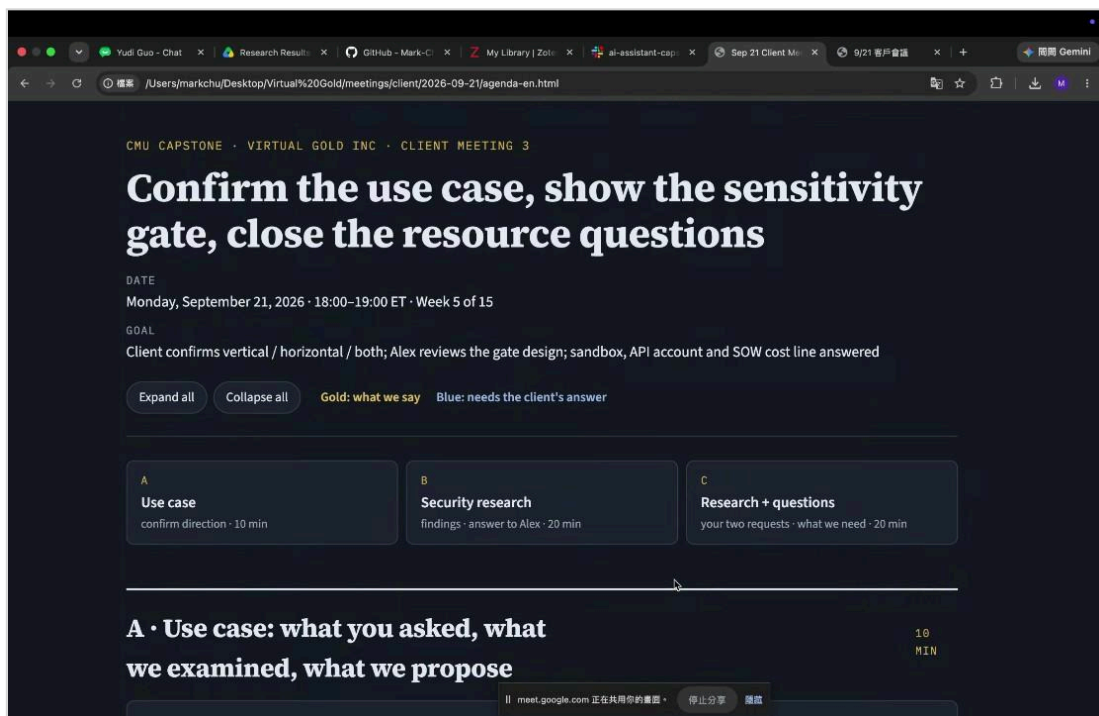
Details

- **Project Labeling and Meeting Agenda:** The meeting opened with Urte Jesina and Mark Chu discussing attendance while waiting for Arvin George and Inderpal Bhandari. Inderpal Bhandari instructed the team to correctly label future meetings as "AI hybrid assistant" rather than "virtual gold AI assistant" to distinguish between the two capstone projects. Mark Chu then outlined the agenda for week five, which included discussing use cases, reviewing security research presented by Yudi, and planning future steps.
- **Vertical Scenario Proposal for Public Filings:** Mark Chu presented a vertical application scenario where a trading or portfolio advice agent would answer user queries based on the latest 10-Q and 10-K public filings from companies. The proposed system would evaluate data confidentiality and model confidence to separate data and answer financial questions. Inderpal Bhandari confirmed after reviewing the documents that using publicly available data makes this vertical scenario very doable and provides strong use case scenarios.
- **Document Access and Sharing Issues:** A discussion occurred regarding document accessibility when Inderpal Bhandari could not initially locate the scope of work PDF for the meeting. Mark Chu explained that the document was attached to today's calendar meeting, but after Inderpal Bhandari reported being unable to find it, Urte Jesina successfully forwarded the agenda document via Gmail.

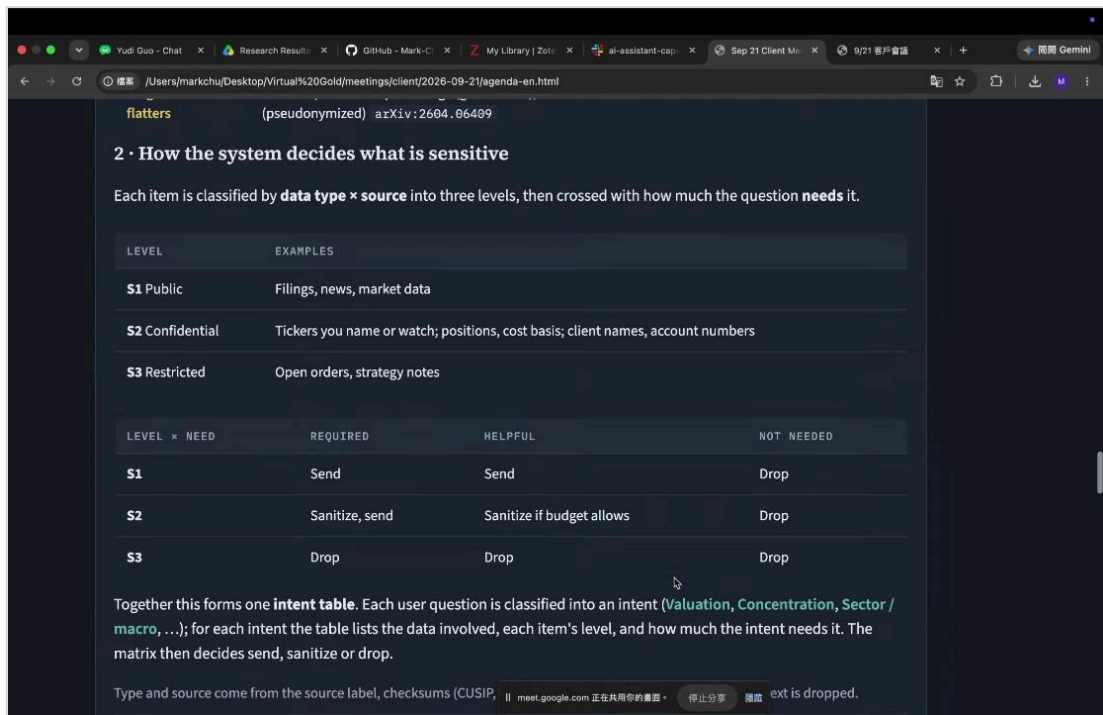
- Horizontal Scenario Proposal and Prioritization:** Mark Chu proposed a horizontal scenario centered on an IT department or customer service Q&A gateway that handles both confidential and open data by masking names, accounts, and phone numbers with placeholders. Inderpal Bhandari and Alex argued that pursuing both horizontal and vertical scenarios simultaneously risked running out of time and recommended staging the horizontal scenario first due to its broader applicability across different company IT departments. Alex and Inderpal Bhandari agreed that focusing on the horizontal module would be the most effective strategy, as it could apply generally before addressing any vertical cases.
- Confidentiality Agreements and Integration Architecture:** Inderpal Bhandari expressed hesitation about sharing their internal architecture due to current limitations in the CMU agreement regarding confidentiality. Inderpal Bhandari noted that they contacted Diane at CMU to add a non-disclosure agreement addendum, expecting a positive response. Alex suggested that if the non-disclosure agreement path fails, the team could build a Model Context Protocol to connect cleanly with the existing open-source OpenClaw architecture used by the VG08 assistant. Mark Chu clarified that their team initially intended to build both horizontal and vertical aspects, but accepted the recommendation to focus on the horizontal scope.



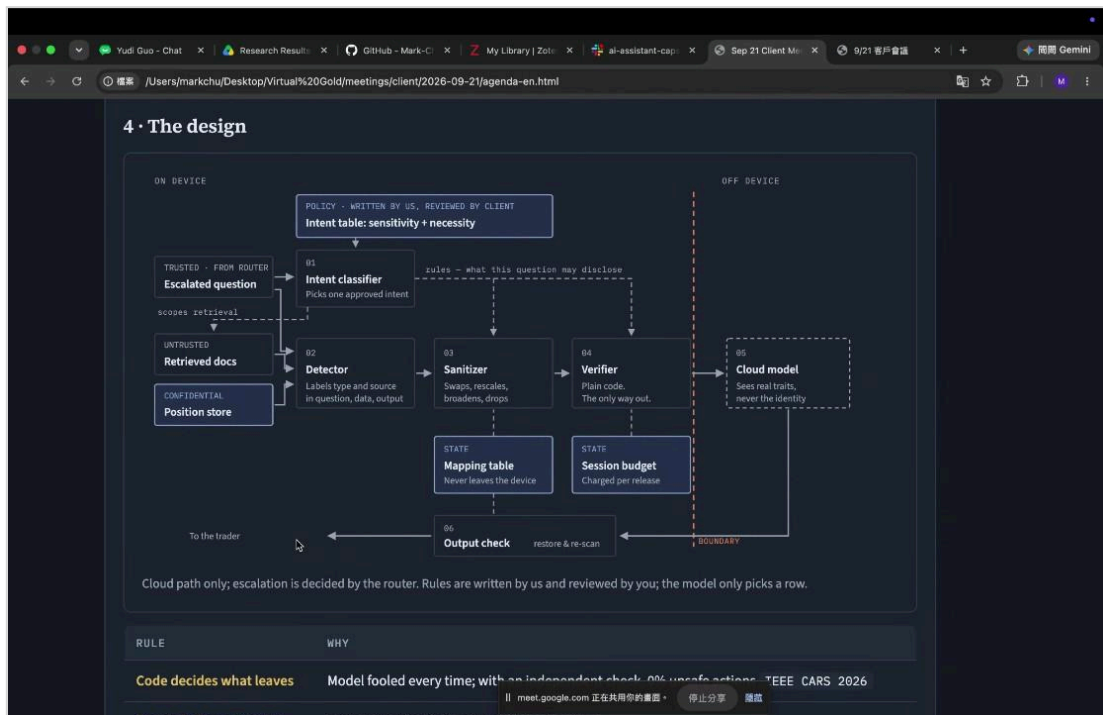
- Re-evaluating Scope as a Chief of Staff Assistant:** Inderpal Bhandari noted that since week five represents roughly one-third of the capstone timeline, splitting focus between horizontal and vertical projects risked completing neither. Anmol Sahu and Mark Chu agreed with this perspective, supporting a primary focus on the horizontal approach. Discussing the specific horizontal focus, Inderpal Bhandari and Alex agreed that framing the scope around a chief of staff assistant was broader and more useful than a strictly IT-department use case, as it would encompass both business and IT issues. Mark Chu confirmed that the team would adjust their research to focus on this horizontal framework.



- Security Research Insights on Sensitive Data and Leakage:** Mark Chu presented security research findings indicating that sensitive data extends beyond personally identifiable information, binary gates are insufficient because only 4% of content actually requires them, re-identification rates reach 9% after detecting identifiers, and conversational single-turn testing understates safety risks because data leakage increases after the first message. To address classification, Chu's research proposed categorizing items across data type and source into three sensitivity levels: public, confidential, and restricted. Furthermore, Mark Chu noted that dealing with sensitive data requires strategies like suppression, generalization, or substitution, which must be reversible to decode sanitized text.



- Security Architecture Components:** Mark Chu outlined a proposed security architecture comprising four main components: an intent classifier using a table of sub-scenarios, an open-source detector to identify sensitive parts in user questions and retrieved documents, a sanitizer to swap or drop sensitive data, and a verifier serving as a safeguard code layer before cloud submission. Alex questioned the implementation details of the pipeline blocks, prompting Mark Chu to clarify that the intent classifier would use a language model while components three and four would use hardcoded rules. Anmol Sahu and Mark Chu acknowledged that this architecture was initially framed around the vertical trading agent and would need to be redesigned to fit the new horizontal scenario.

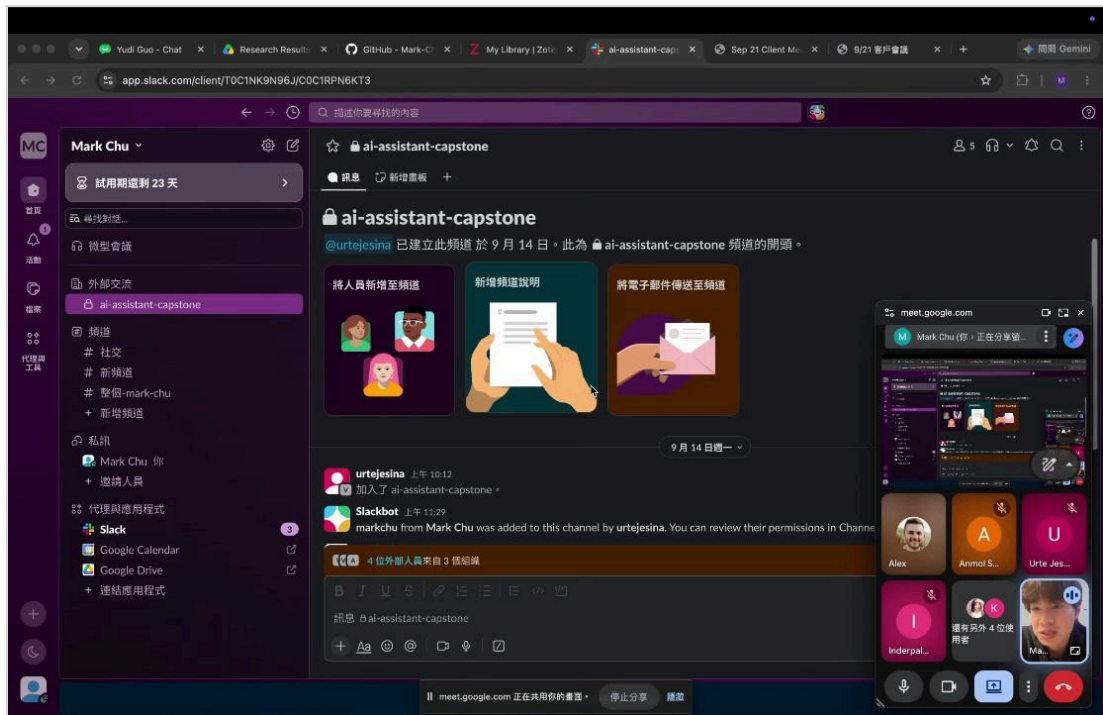


- Introduction of Aarvin George and VG Assistant 08 Architecture:**

Inderpal Bhandari introduced Aarvin George, an alumnus and BG developer based in Pittsburgh, who presented an overview of the enterprise assistant project, currently designated as BG assistant 08. Aarvin George explained that while waiting for the non-disclosure agreement resolution, the hybrid local model capstone team should build a standalone server with an HTTP endpoint rather than a Model Context Protocol server, allowing the team to test prompts independently. Aarvin George specified that this server would act as an intermediate layer to receive, sanitize, and decompose instructions before passing them to the main agent orchestration. Alex and Anmol Sahu agreed to proceed with designing this API endpoint integration.
- Ground Truth Conditions and Testing Frameworks:**

Aarvin George advised the capstone team to establish clear ground truth conditions, success metrics, and regression testing protocols to maintain a measured development approach without building in the dark. Anmol Sahu proposed adding an evaluation layer using a language model to verify responses against a confidence threshold before sending them back. Aarvin George noted that while language model evaluation is a good approach, human involvement is still required to define correct conditions, and promised to provide an initial API contract by the following day while noting that contracts would likely evolve over the first week and a half.

- Administrative Logistics for Google Drive and Slack:** Mark Chu raised logistical questions near the end of the meeting regarding workspace and communication tools. Inderpal Bhandari confirmed that the team should use the Virtual Gold Google Drive and invite Aarvin George, Alex, Inderpal Bhandari, and Alra. Mark Chu and Anmol Sahu highlighted an issue regarding the Slack workspace showing a 30-day trial expiration on their accounts, which Urte Jesina and Inderpal Bhandari noted could be managed or set up with Alra's assistance.



- Slack Communication Platform Setup:** The team addressed a platform limitation where merging a single Slack group with two capstones as separate threads required a pro feature account. Urte Jesina suggested switching to a different platform if necessary, while Alex proposed creating a separate Slack group per capstone to bypass the limitation. Mark Chu agreed to test this solution, send the link to everyone, and follow up via email or in the next meeting if problems persisted.
- Architecture and System Diagram Deliverables:** The team discussed preparations for the next meeting, focusing on building a comprehensive architecture and system diagram for the horizontal plot. Inderpal Bhandari advised Mark Chu to create an initial draft quickly to accelerate progress, noting that a preliminary version did not need to be perfect. Inderpal Bhandari additionally stated that Aarvin George's prepared architecture would be shared soon to help them match things up. Mark Chu confirmed

they would focus on the horizontal scenario, incorporate advice and concepts from Aarvin George, and update a naming convention to "hybrid". Inderpal Bhandari agreed that this approach accurately captured the required deliverables for the next discussion.

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